

1 **Projection-Based Safety Shielding for Provably Direction-Compliant COLREGs Collision Avoidance**
2 **in Continuous Action Spaces**

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1 ABSTRACT

2 **Objectives.** Autonomous surface vessels must obey the International Regulations for Preventing Collisions
3 at Sea (COLREGs). The give-way rules are directional. A track that avoids a collision but passes on
4 the wrong side is still a violation. This paper builds a per-step guarantee of directional compliance for
5 continuous control commands.

6 **Methods.** The give-way clause is written as a half-plane constraint on the plane of control commands.
7 It is intersected with the actuator limits and a one-step collision-free condition, which gives a convex safe
8 control set. A small quadratic program projects the command of the policy onto the nearest point of that
9 set. The projection sits inside the environment transition, so the policy learns under its own corrected
10 commands. The data are 2,000 simulated two-vessel encounters, split into 1,400 training and 600 test cases.
11 Nine configurations form a single-variable ladder, each trained with eight fixed seeds.

12 **Findings.** On every give-way step where the projection is feasible, directional compliance holds by
13 construction. It does not depend on the training outcome or the seed. Steps that switch to the fallback
14 branch are not covered. On the test set, the violation count per episode is [TBD] and the collision rate is
15 [TBD]. The ablation gains are significant under a paired signed-rank test. The arrival rate is not claimed as
16 a benefit.

17 **Novelty.** To the best of the authors' knowledge, this is the first projection-based safety mechanism that
18 enforces directional COLREGs compliance on continuous control commands. The paper also states what is
19 proved and what is not.

20 **Practical Applications.** The compliance check is independent of training. The mechanism can sit above
21 an existing controller as a compliance layer for functional verification and liability assessment. The online
22 cost is one small quadratic program per step, so it runs in real time without special hardware.

23 **Keywords:** COLREGs compliance, safe reinforcement learning, projection-based safety shield, continuous
24 action space, quadratic programming, unmanned surface vehicle

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